

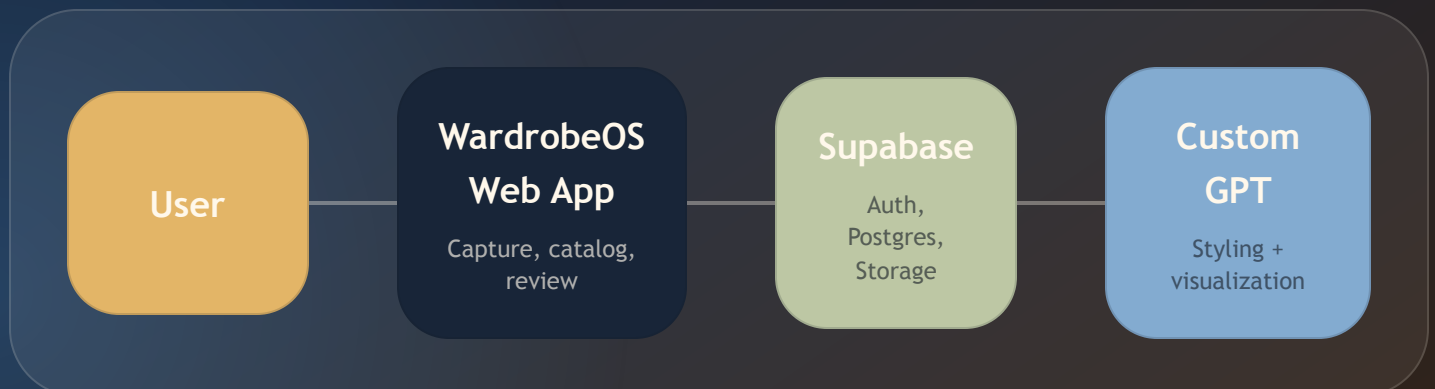
Closet Data Meets AI Styling

PRIVATE BETA

Architecture Edition

WEB APP + BACKEND + GPT
ACTIONS

A compact architecture tour of the web app, backend, Supabase data layer, Gemini tagging, iPhone PWA, and ChatGPT companion that turns a real closet into a personal styling system.



The Product Boundary

WardrobeOS is deliberately split into two surfaces. The web app is the system of record: accounts, closet uploads, metadata, saved outfits, preferences, and operational workflows. The Custom GPT is the creative companion: conversation, outfit planning, and visual styling.

This separation keeps the app practical and affordable. Durable data stays in WardrobeOS. Expensive creative image generation happens in ChatGPT, using the user's own ChatGPT plan.

“The system of record stays boring. The styling surface gets expressive.”

WardrobeOS Web App

Login and profile

Single and bulk upload

AI tag review

Closet management

Outfits, packing, insights

Custom GPT

Reads closet via OAuth

Conversational styling

Outfit ideation

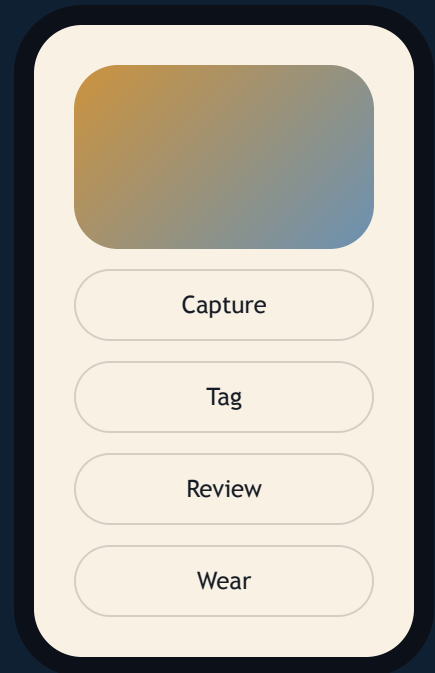
Visualization briefs

ChatGPT image generation

Frontend And Experience

A fast wardrobe cockpit

The frontend uses Next.js App Router with a PWA-friendly shell. It is optimized for capture, review, and decision-making rather than generic inventory management.



Dashboard

Wear logging, recent activity, quick closet health signals.

Closet

Search, filter, list/grid views, item detail and archival.

Bulk Upload

Upload up to 50 photos, apply AI tags, review before save.

Outfit Lab

Explainable suggestions scored by color, occasion, weather, and rotation.

Buy Next

Simulates the next purchase that unlocks the most outfits.

PWA

iPhone install path for quick photo capture and email-code login.

Backend, Data, And Storage

FRONTEND

Next.js App Router, TypeScript, Tailwind, PWA shell, Zustand interaction state.

BACKEND

Vercel Route Handlers for auth boundaries, validation, AI calls, uploads, and GPT Actions.

DATA

Supabase Auth, Postgres, RLS, Storage, wardrobe tables, image metadata, OAuth grant tables.

AI LAYER

Gemini for reviewed item tagging. ChatGPT Custom GPT for styling conversation and visual previews.

Route handlers as the server boundary

`/api/bootstrap`

Hydrates signed-in profile, default closet, and wardrobe items.

`/api/items/*`

Create, update, archive, mark worn, upload and confirm item images.

`/api/ai/*`

Single and bulk Gemini tagging behind user review.

`/api/gpt/*`

OAuth and read-only GPT Actions for closet styling.

profiles

closets

items

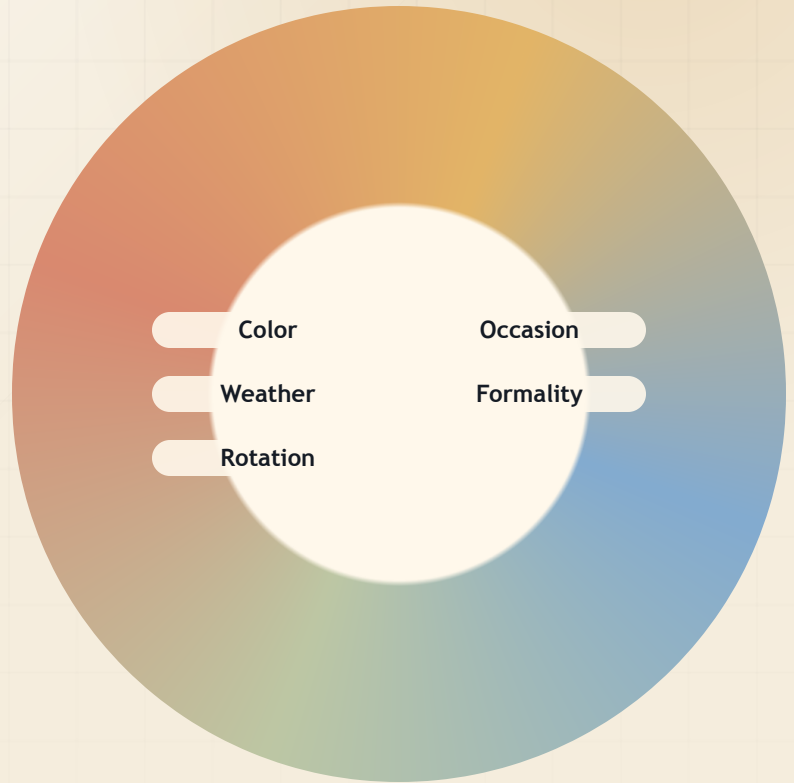
item_images

saved_outfits

gpt_oauth_tokens

Deterministic Intelligence

The recommendation engine is intentionally deterministic. Color, occasion, season, weather, formality, and rotation logic live in pure domain modules. That makes every recommendation explainable and testable.



AI is invited into the workflow only where it helps: reading item photos, producing practical tags, and helping ChatGPT communicate styling ideas. It is not the ranking source of truth.

Design principle: AI suggestions are reviewed. Scoring rules are inspectable. User data remains structured.

AI Tagging And Bulk Intake

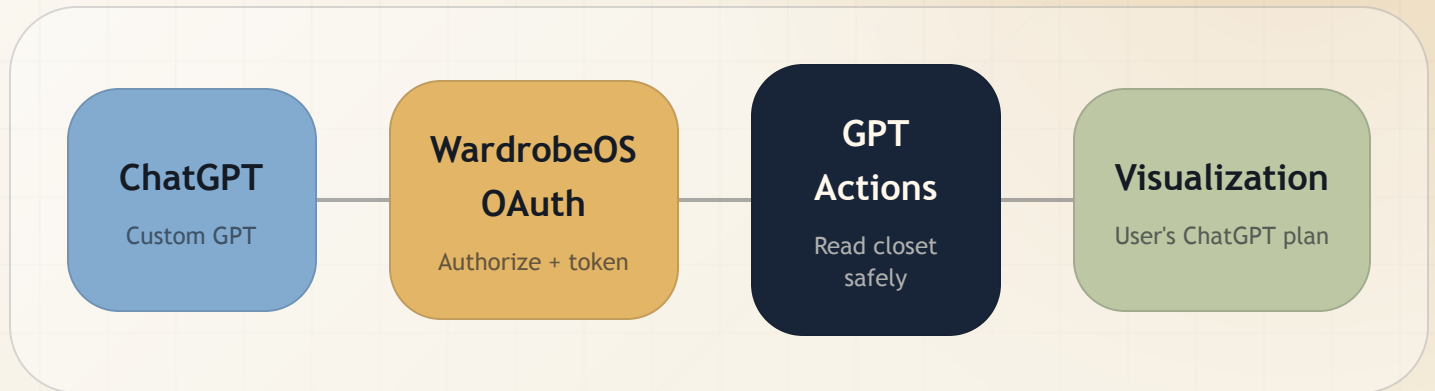


The model currently used for tagging is Gemini 3.1 Flash Lite Preview. It was chosen after direct comparison testing because bulk tagging quality held up while latency and free-tier fit improved.

The bulk flow is the data-collection wedge: many item photos can be processed at once, but users keep final control through a review queue.



ChatGPT Companion

`getWardrobeProfile``listClosetItems``getClosetItem``listSavedOutfits``suggestClosetOutfits``createVisualizationBrief`

ChatGPT receives a WardrobeOS-scoped bearer token, not Supabase credentials. Refresh tokens are hashed in Supabase. Full-body photos are uploaded directly to ChatGPT for visualization, not stored by WardrobeOS.

Deployment And Quality

Production

Vercel hosts the Next.js app, route handlers, PWA assets, and GPT OpenAPI schema.

Data Plane

Supabase owns Auth, Postgres, RLS, and item image storage.

AI Services

Gemini handles reviewed tagging. ChatGPT handles companion conversation and image visualization.

Lint

Typecheck

Unit Tests

Build

Playwright

The architecture creates room for better data capture, better wardrobe intelligence, and richer AI styling while keeping WardrobeOS focused on structured closet truth.

